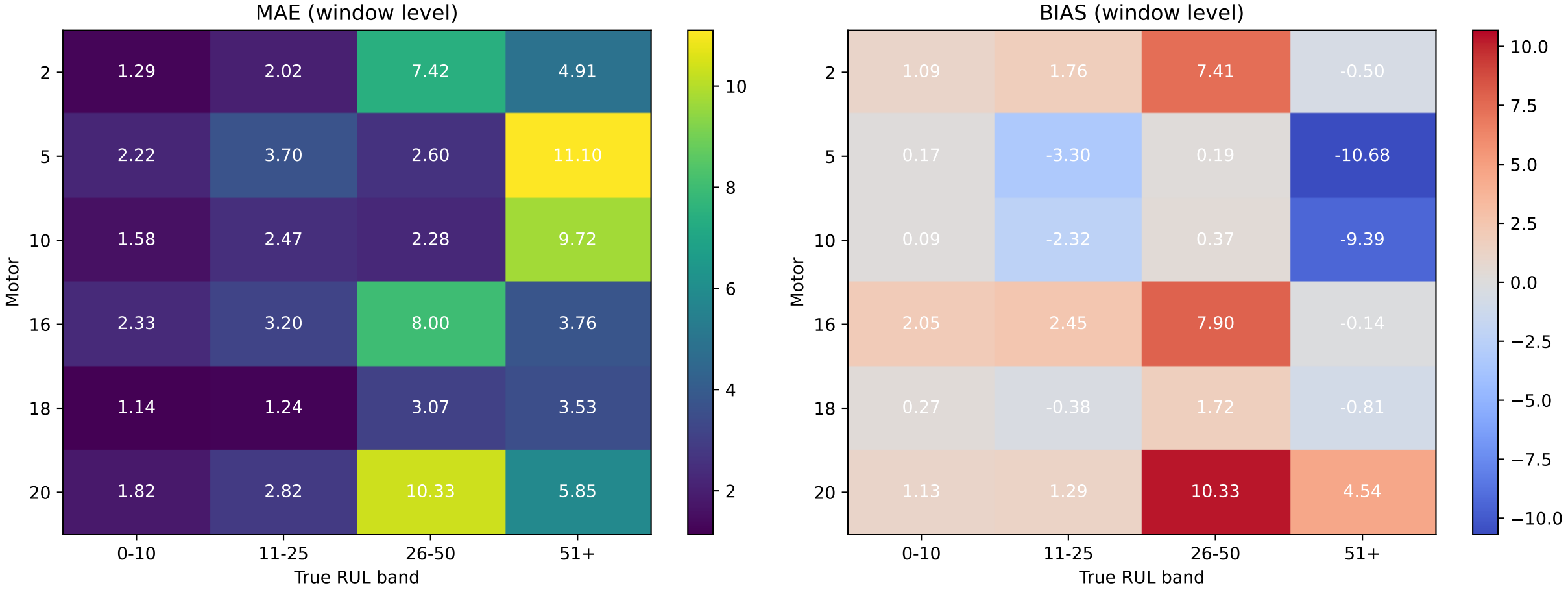

MAE (window level)
Motor
2
5
10
16
18
20
1.29
2.02
7.42
4.91
2.22
3.70
2.60
11.10
1.58
2.47
2.28
9.72
2.33
3.20
8.00
3.76
1.14
1.24
3.07
3.53
1.82
2.82
10.33
5.85
0-10
11-25
26-50
51+
True RUL band
10
8
6
4
2
BIAS (window level)
Motor
2
5
10
16
18
20
1.09
1.76
7.41
-0.50
0.17
-3.30
0.19
-10.68
0.09
-2.32
0.37
-9.39
2.05
2.45
7.90
-0.14
0.27
-0.38
1.72
-0.81
1.13
1.29
10.33
4.54
0-10
11-25
26-50
51+
True RUL band
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0

Motor 2: matched prediction windows, averaged per flight

RUL cycles

True RUL

Mean prediction

Error cycles

Mean signed error

Mean absolute window error

Conditions (within-motor z)

alt

Mach

TRA

T2

Health modifier

fan_eff_mod

fan_flow_mod

LPC_eff_mod

LPC_flow_mod

HPC_eff_mod

HPC_flow_mod

HPT_eff_mod

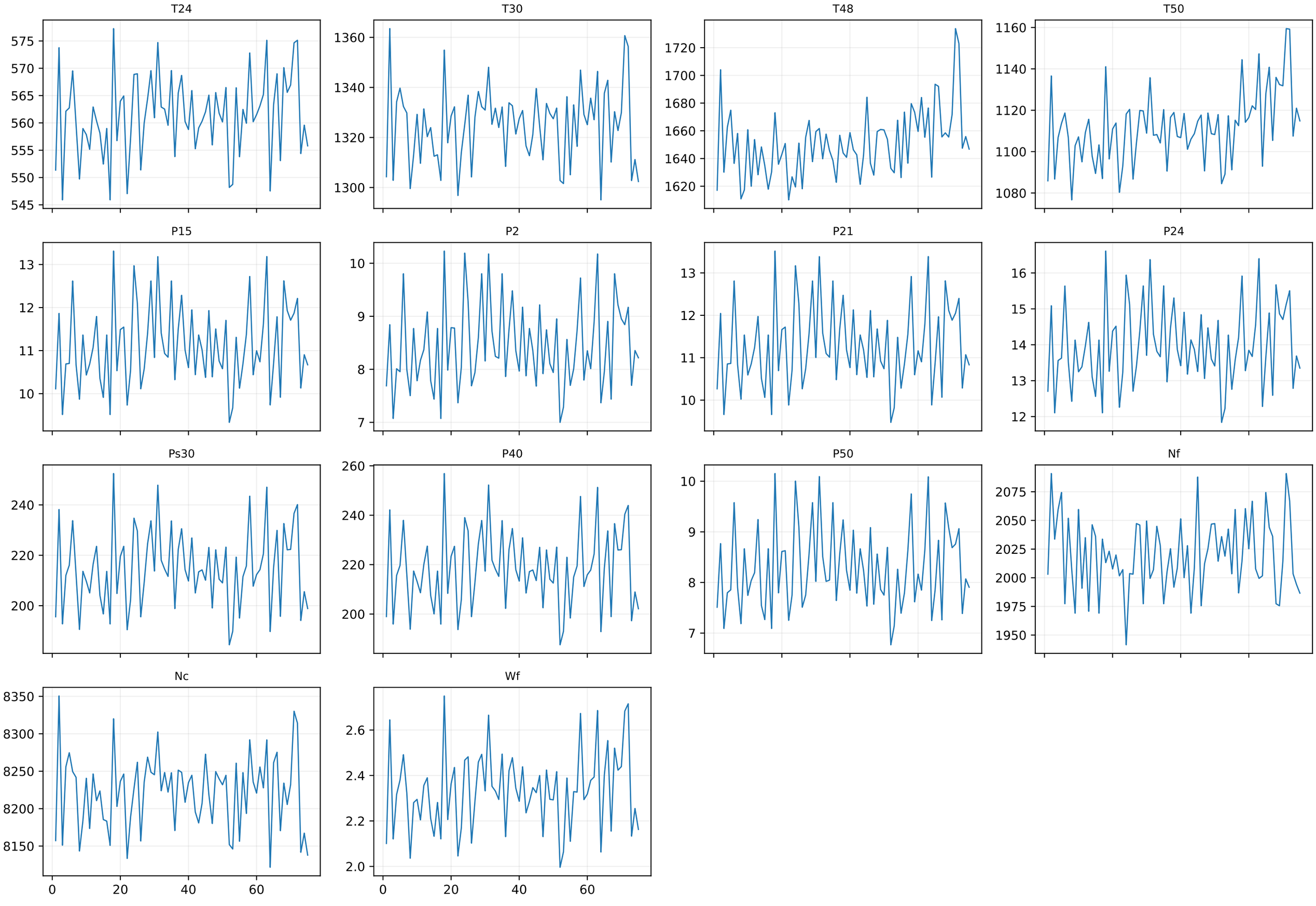
HPT_flow_mod

LPT_eff_mod

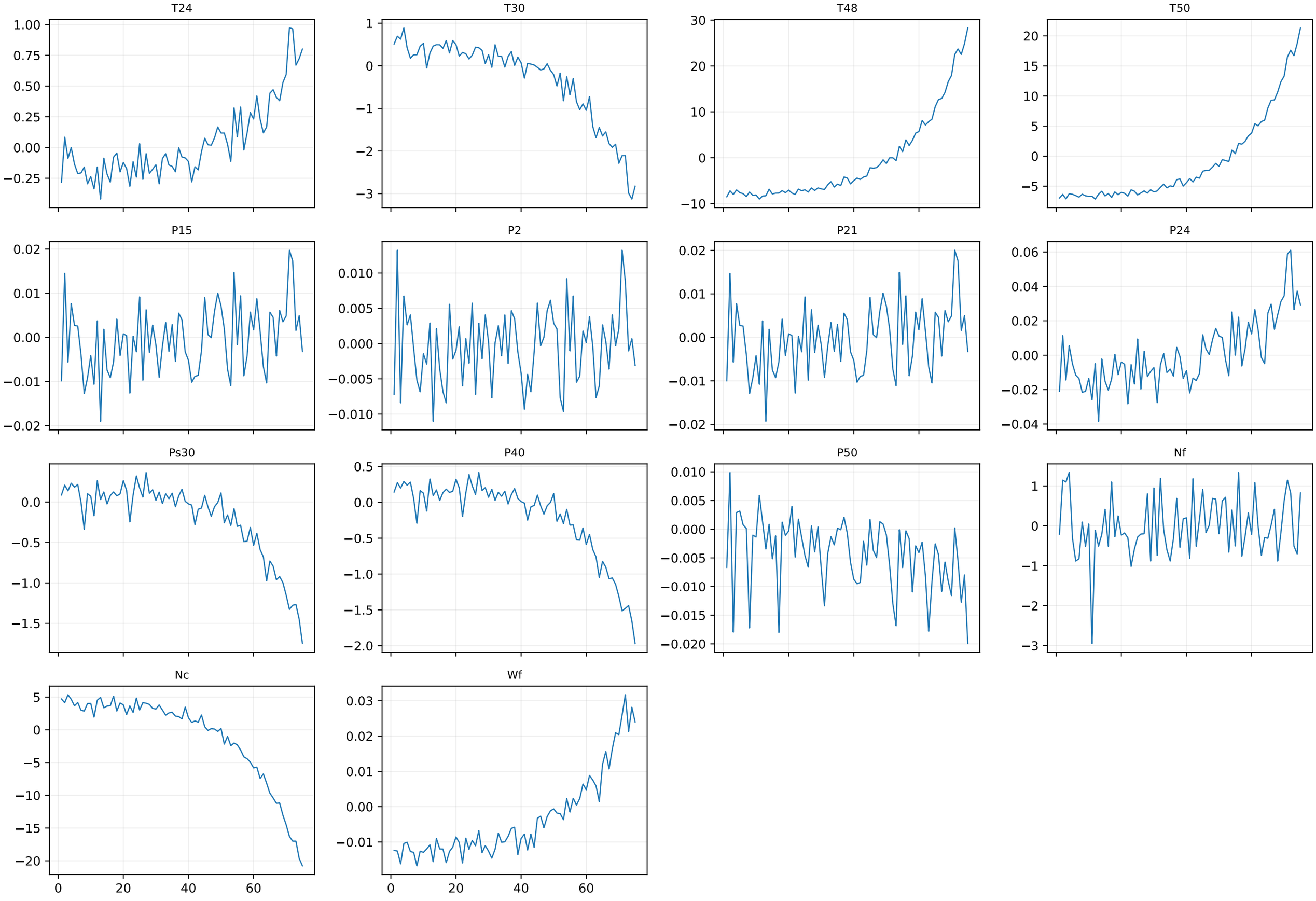
LPT_flow_mod

Flight cycle

Motor 2: sensors; flight means of selected windows



Motor 2: condition_residuals; flight means of selected windows



Motor 5: matched prediction windows, averaged per flight

RUL cycles

True RUL

Mean prediction

Error cycles

Mean signed error

Mean absolute window error

Conditions (within-motor z)

alt

Mach

TRA

T2

Health modifier

fan_eff_mod

fan_flow_mod

LPC_eff_mod

LPC_flow_mod

HPC_eff_mod

HPC_flow_mod

HPT_eff_mod

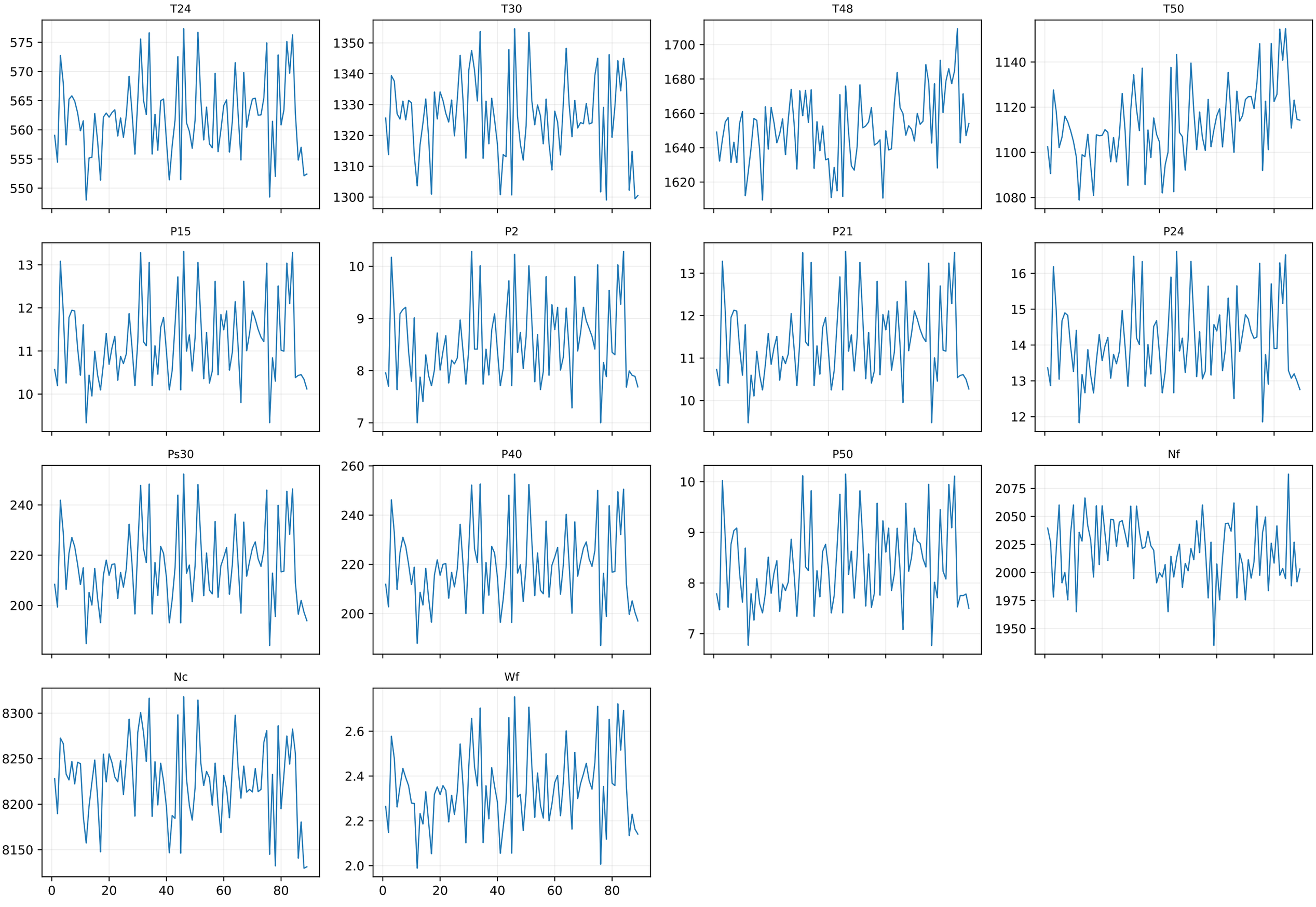
HPT_flow_mod

LPT_eff_mod

LPT_flow_mod

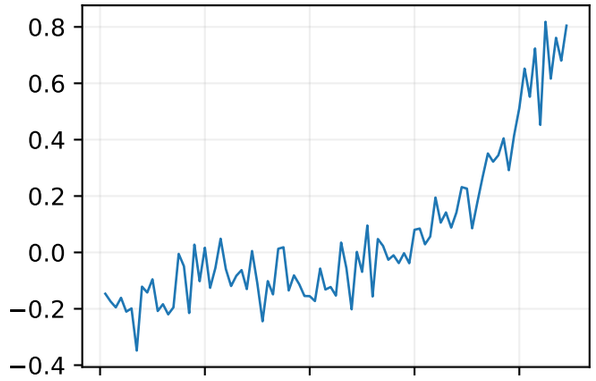
Flight cycle

Motor 5: sensors; flight means of selected windows

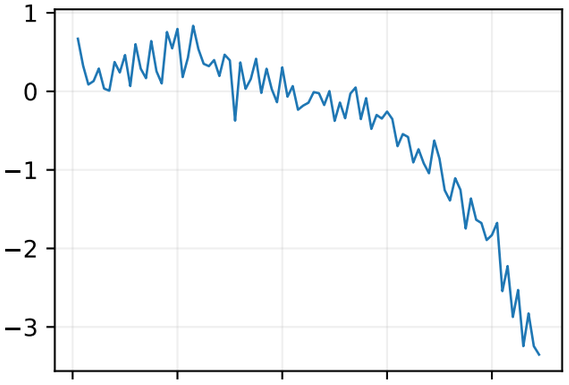


Motor 5: condition_residuals; flight means of selected windows

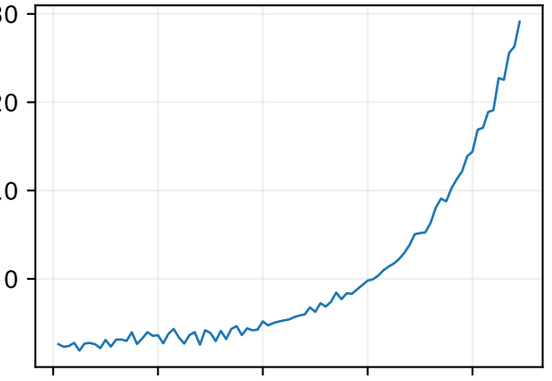
T24



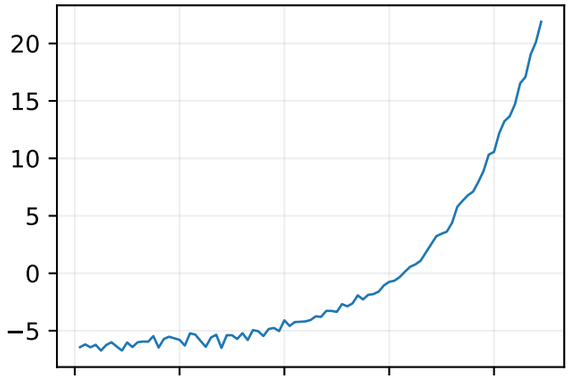
T30



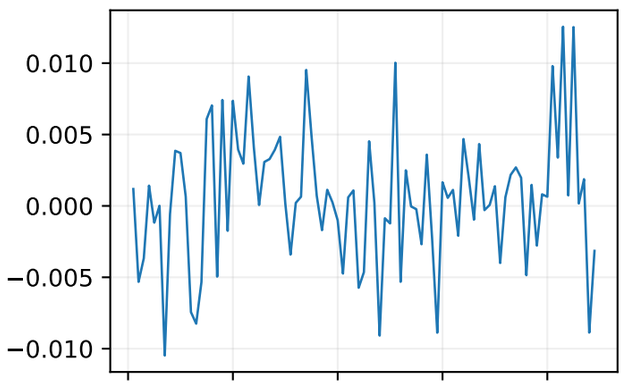
T48



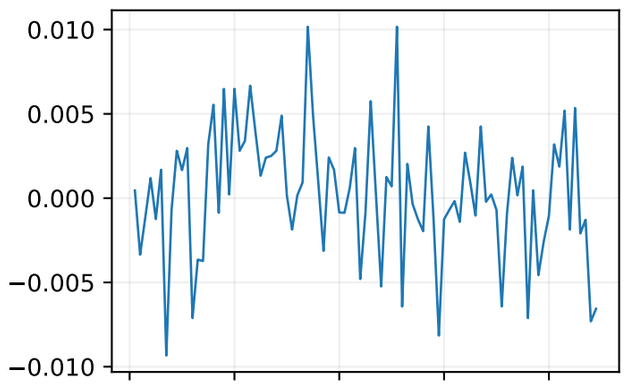
T50



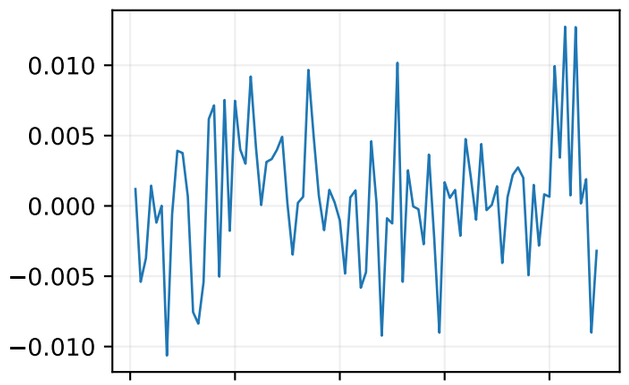
P15



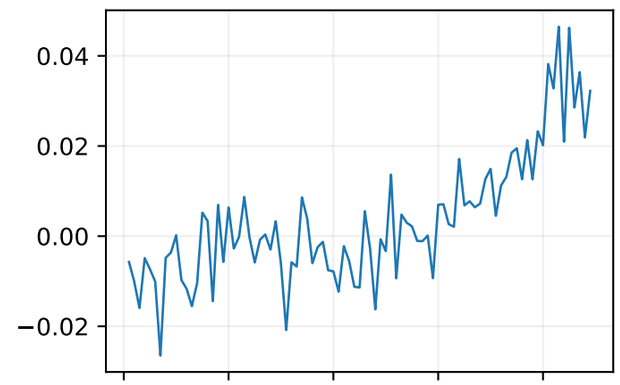
P2



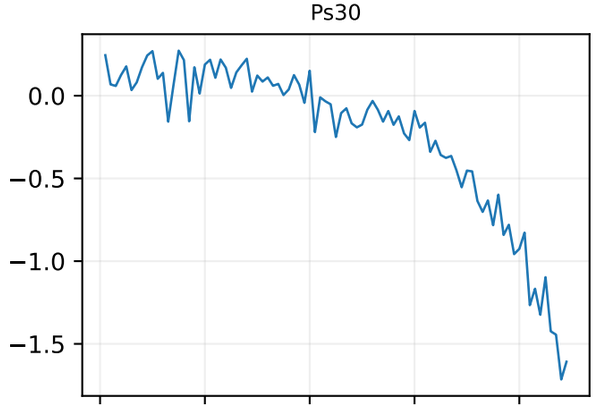
P21



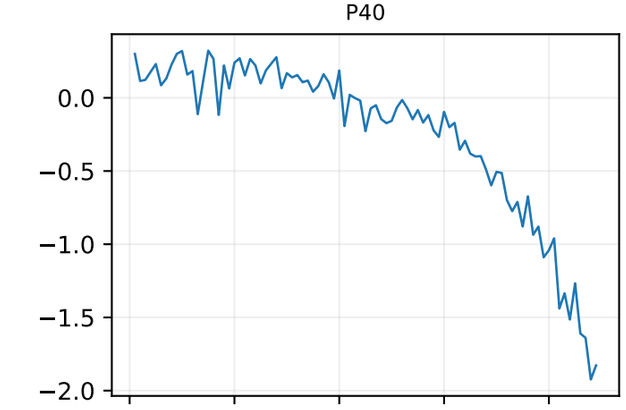
P24



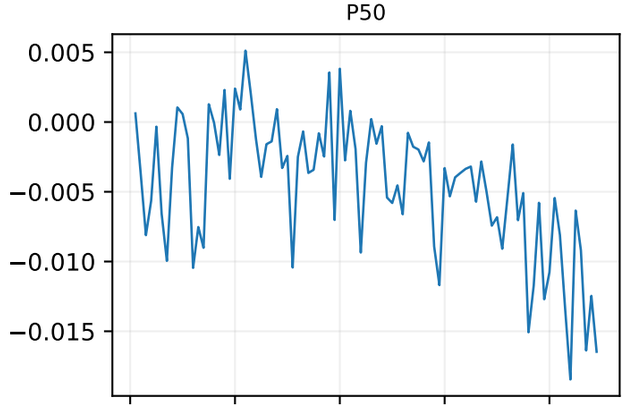
Ps30



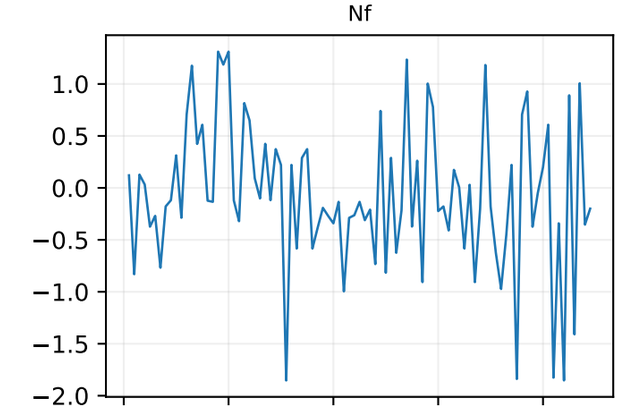
P40



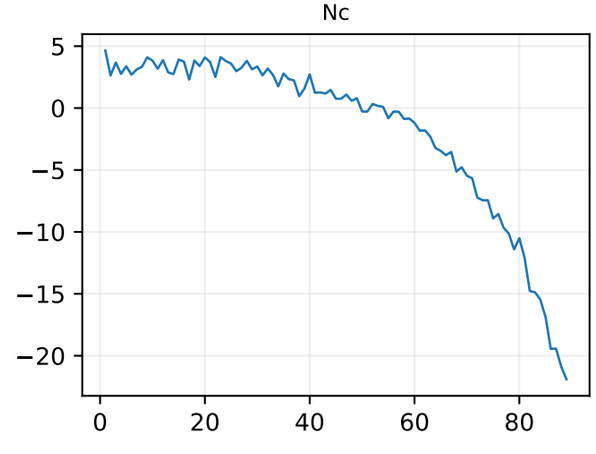
P50



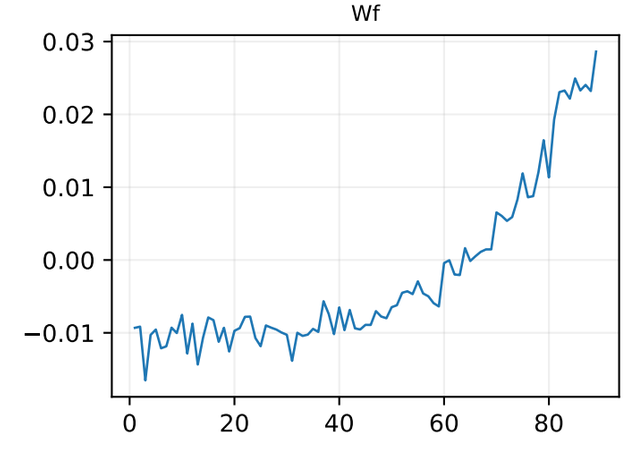
Nf



Nc



Wf



Flight cycle

Motor 10: matched prediction windows, averaged per flight

RUL cycles

True RUL

Mean prediction

Error cycles

Mean signed error

Mean absolute window error

Conditions (within-motor z)

alt

Mach

TRA

T2

Health modifier

fan_eff_mod

fan_flow_mod

LPC_eff_mod

LPC_flow_mod

HPC_eff_mod

HPC_flow_mod

HPT_eff_mod

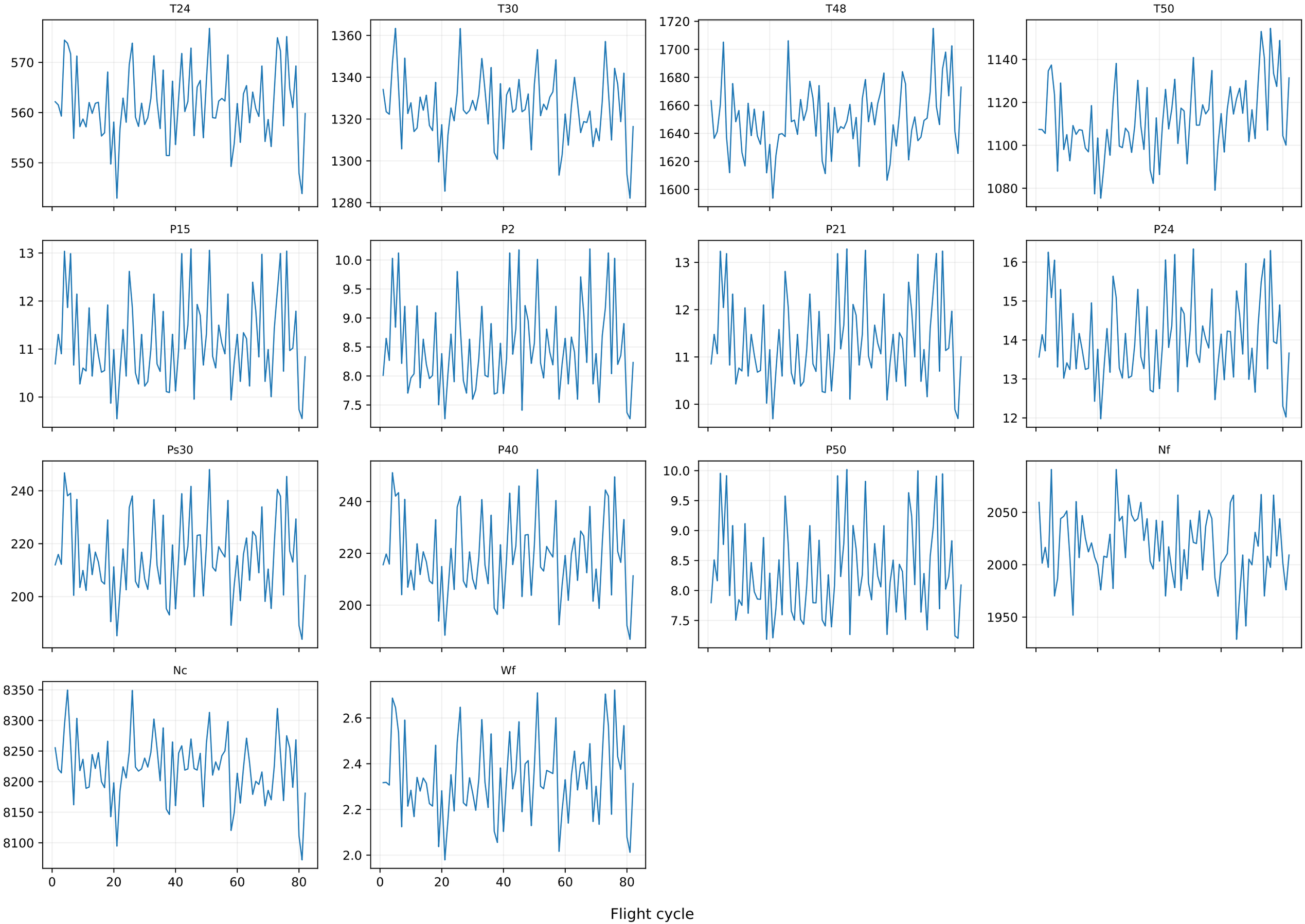
HPT_flow_mod

LPT_eff_mod

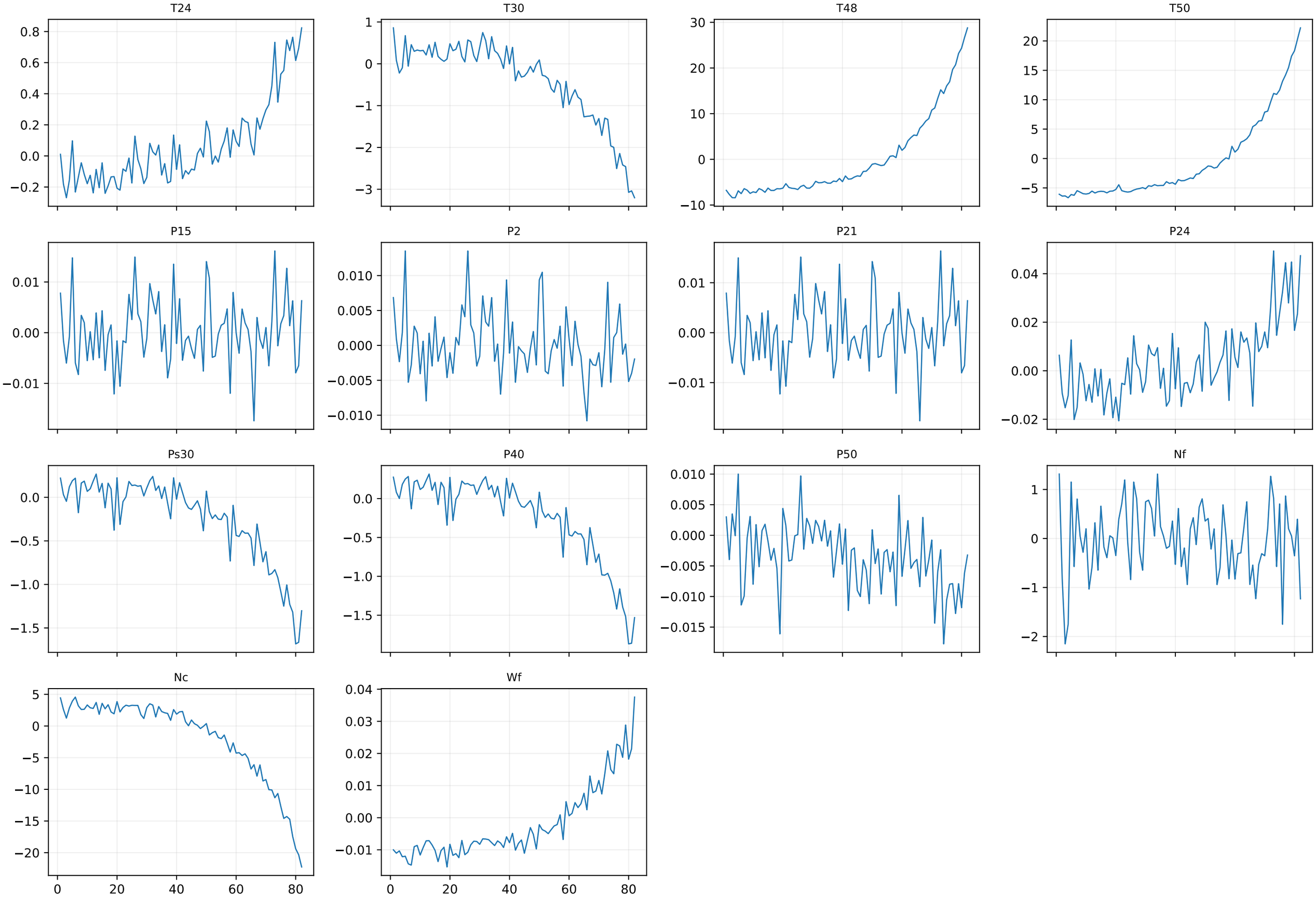
LPT_flow_mod

Flight cycle

Motor 10: sensors; flight means of selected windows



Motor 10: condition_residuals; flight means of selected windows



Motor 16: matched prediction windows, averaged per flight

RUL cycles

True RUL

Mean prediction

Error cycles

Mean signed error

Mean absolute window error

Conditions (within-motor z)

alt

Mach

TRA

T2

Health modifier

fan_eff_mod

fan_flow_mod

LPC_eff_mod

LPC_flow_mod

HPC_eff_mod

HPC_flow_mod

HPT_eff_mod

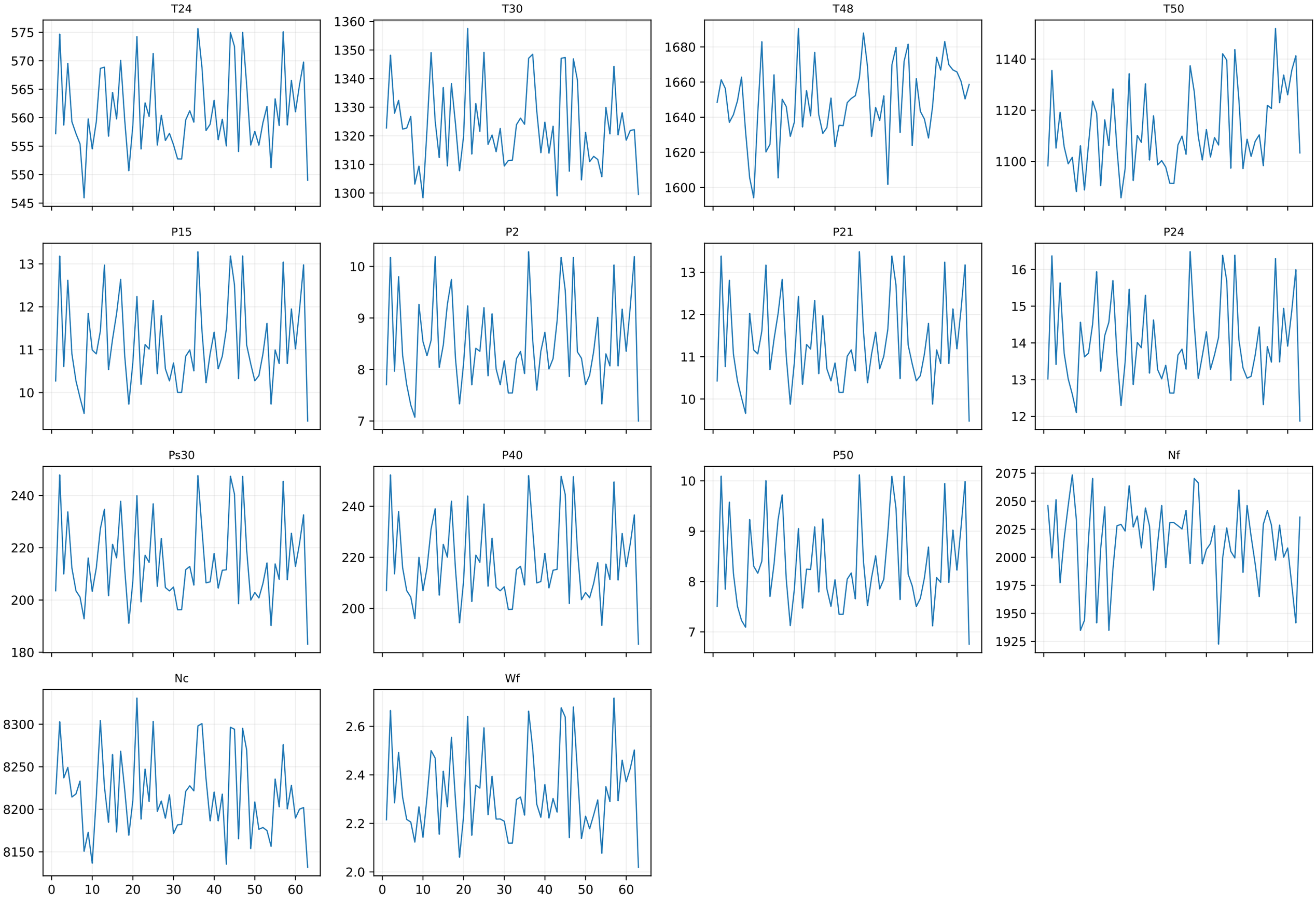
HPT_flow_mod

LPT_eff_mod

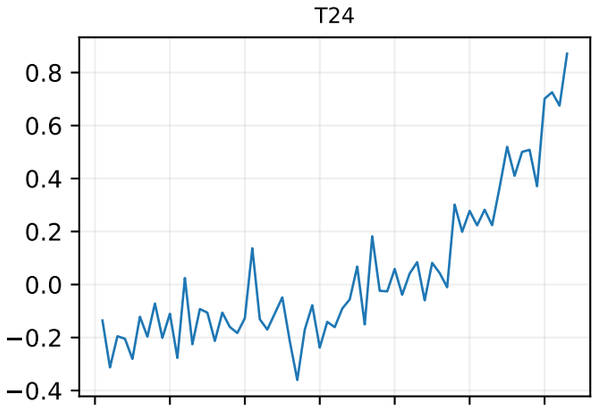
LPT_flow_mod

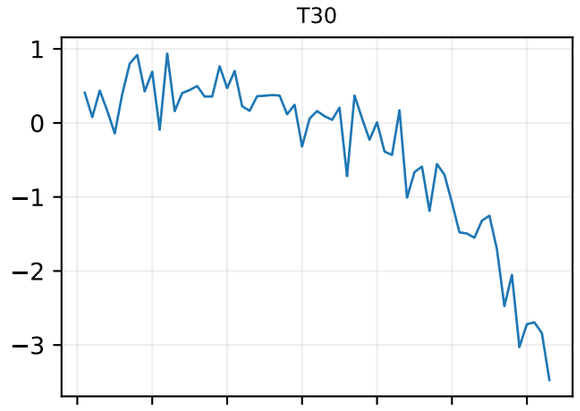
Flight cycle

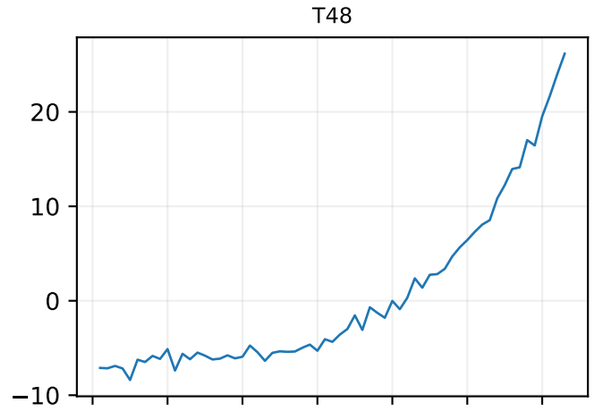
Motor 16: sensors; flight means of selected windows

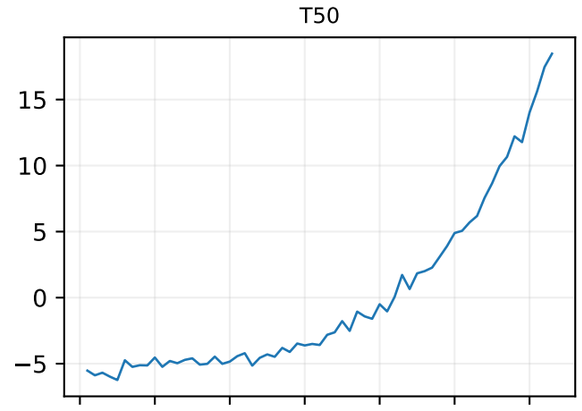


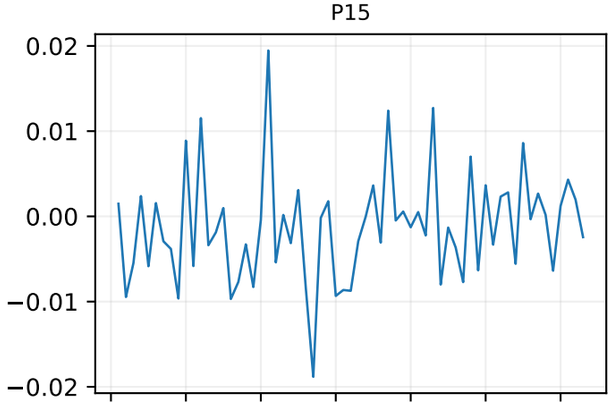
Motor 16: condition_residuals; flight means of selected windows

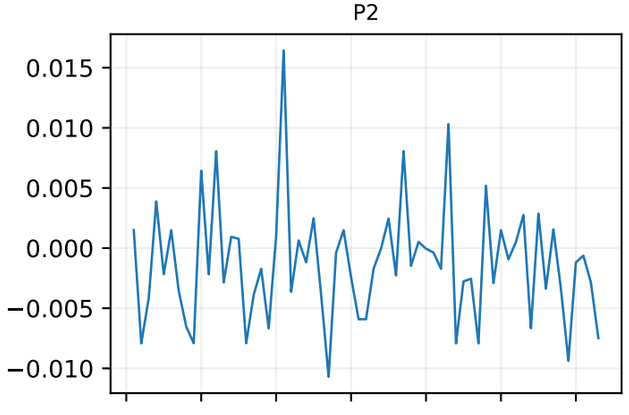


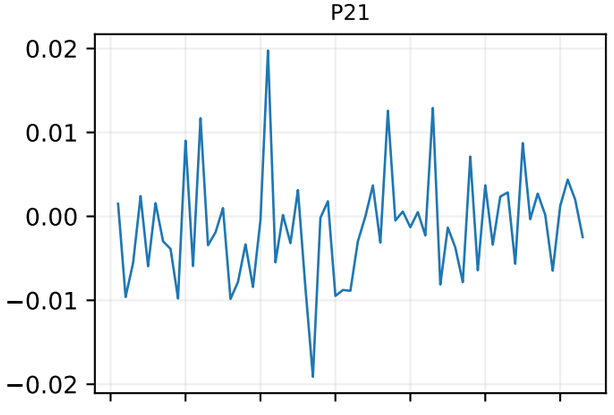


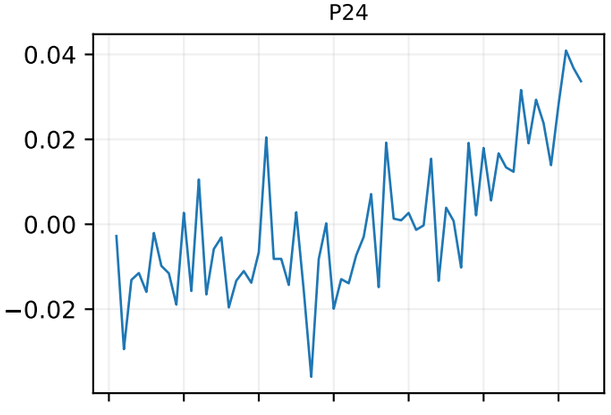


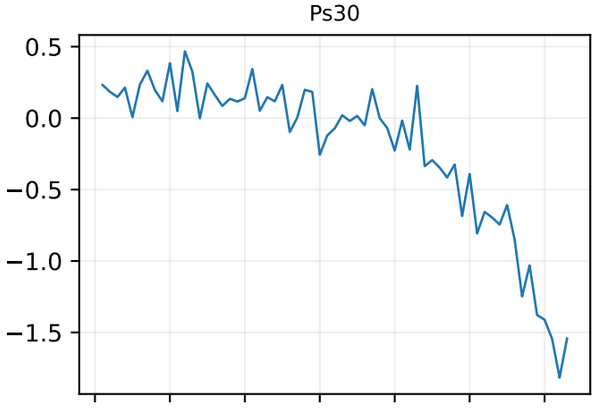


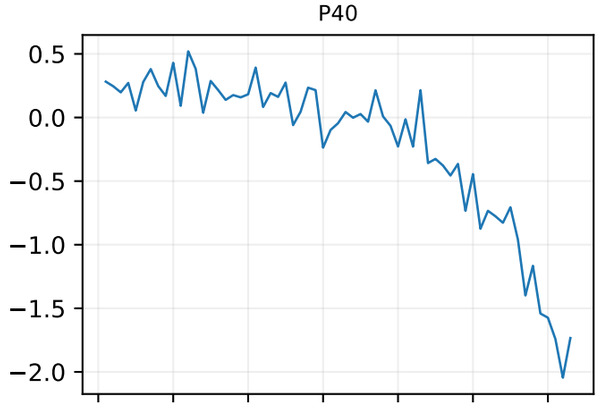


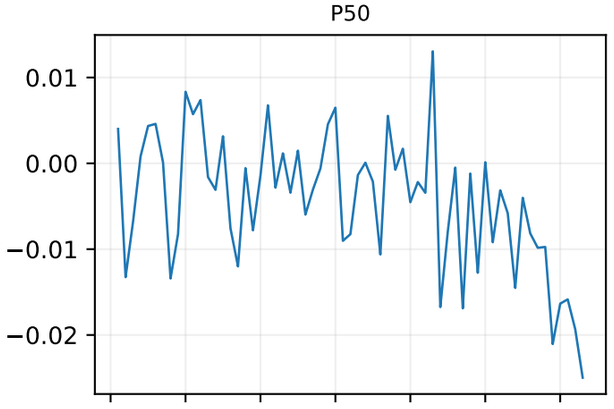


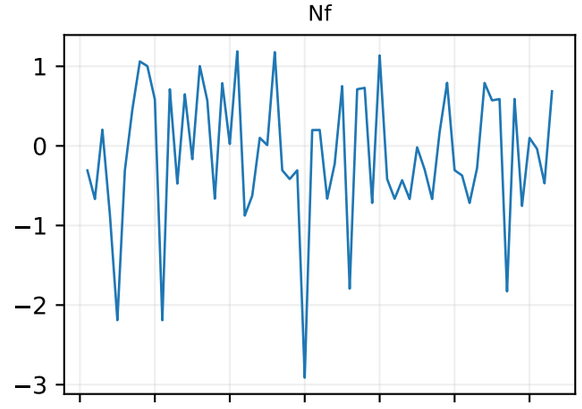


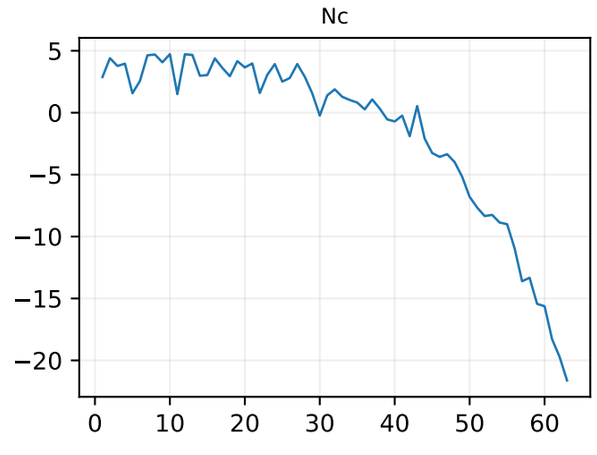


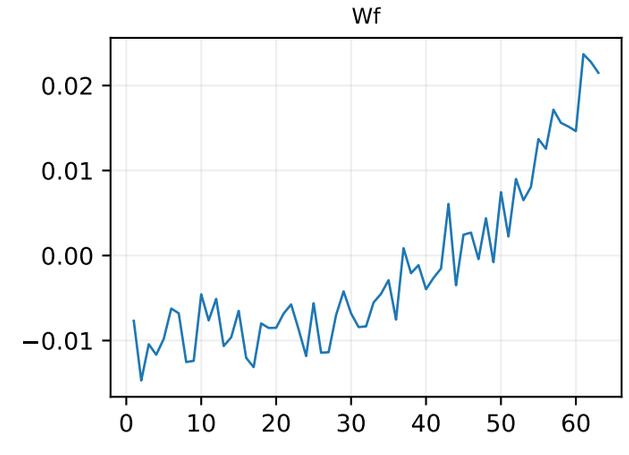












Flight cycle

Motor 18: matched prediction windows, averaged per flight

RUL cycles

True RUL

Mean prediction

Error cycles

Mean signed error

Mean absolute window error

Conditions (within-motor z)

alt

Mach

TRA

T2

Health modifier

fan_eff_mod

fan_flow_mod

LPC_eff_mod

LPC_flow_mod

HPC_eff_mod

HPC_flow_mod

HPT_eff_mod

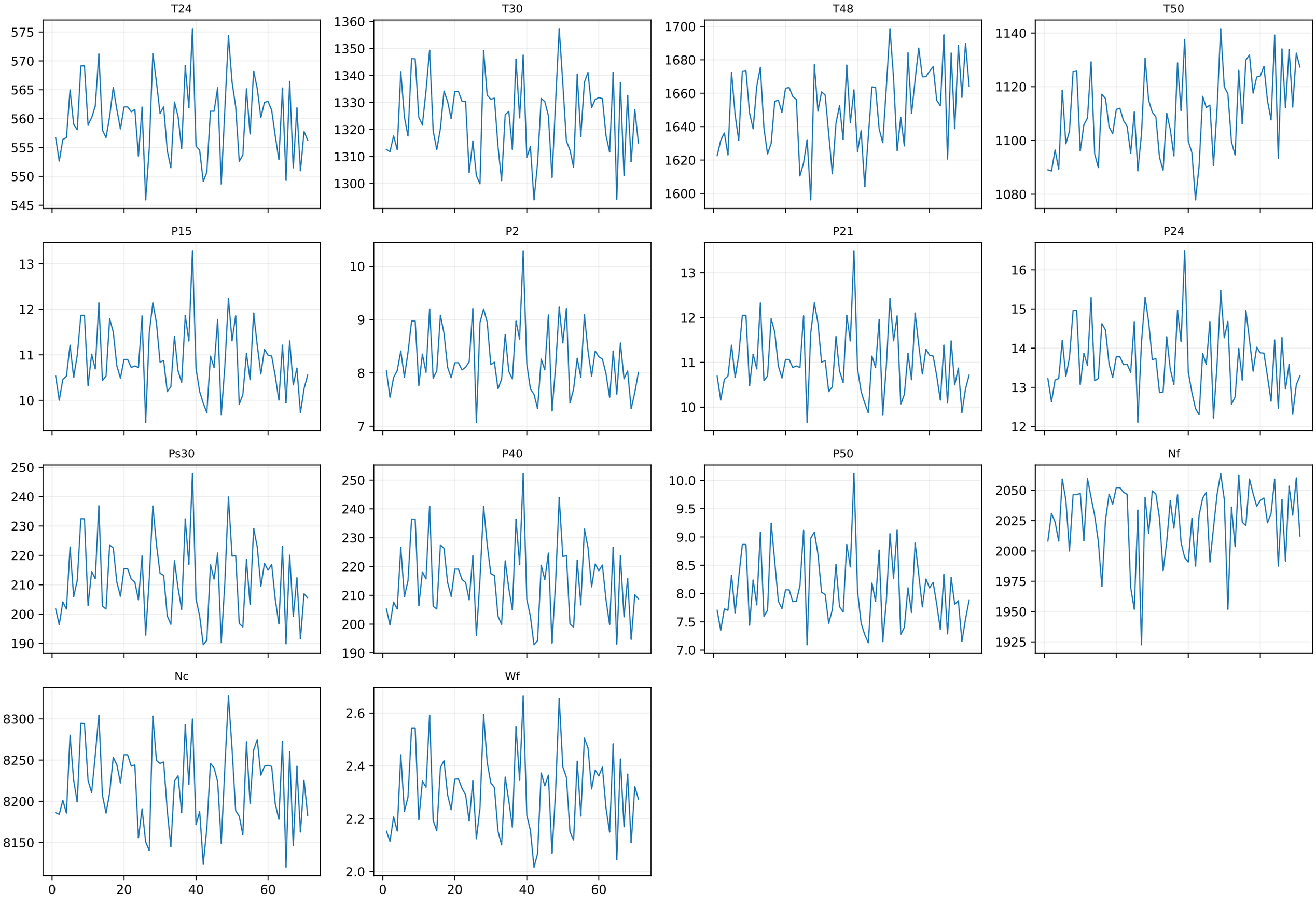
HPT_flow_mod

LPT_eff_mod

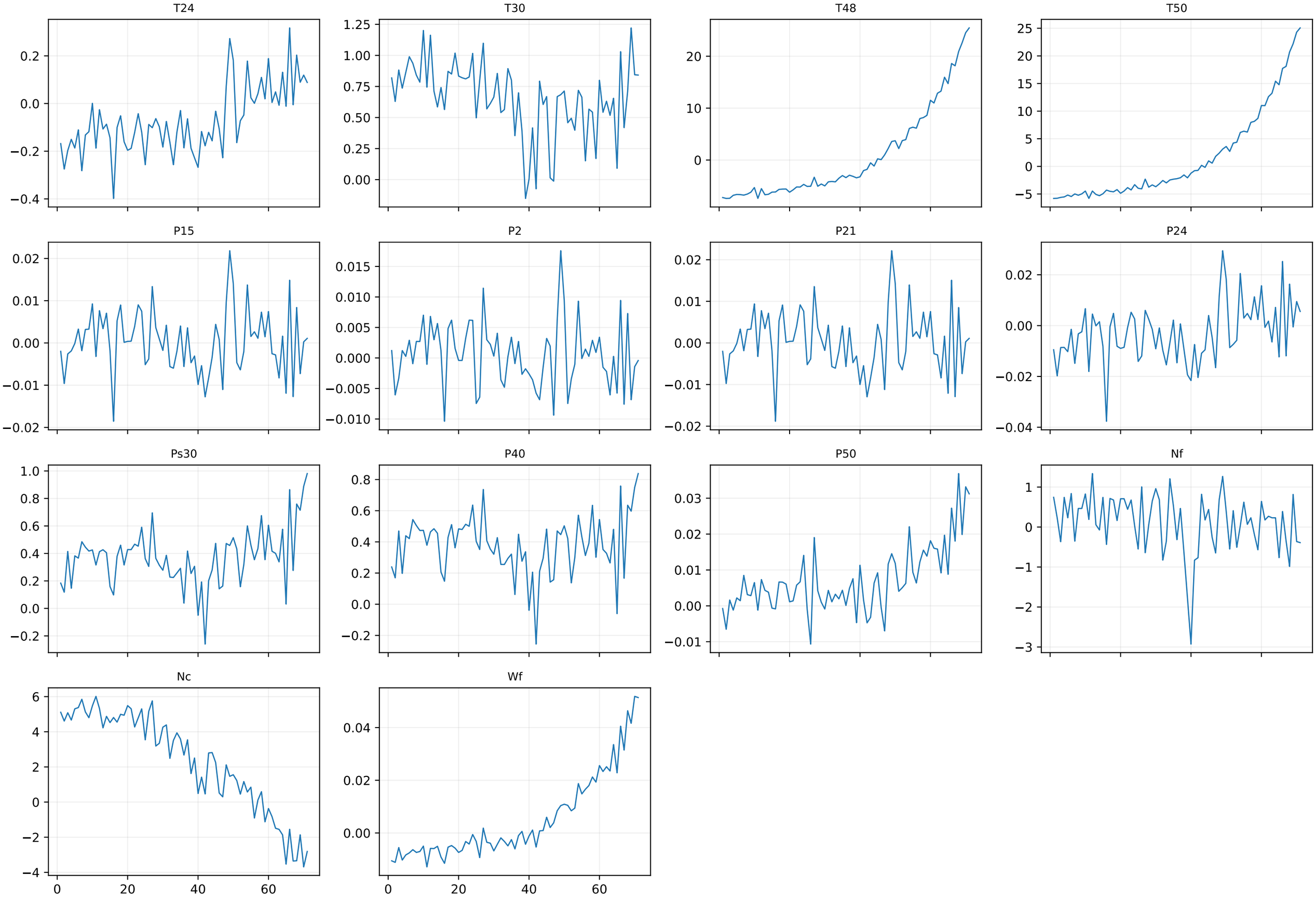
LPT_flow_mod

Flight cycle

Motor 18: sensors; flight means of selected windows



Motor 18: condition_residuals; flight means of selected windows



Motor 20: matched prediction windows, averaged per flight

RUL cycles

True RUL

Mean prediction

Error cycles

Mean signed error

Mean absolute window error

Conditions (within-motor z)

alt

Mach

TRA

T2

Health modifier

fan_eff_mod

fan_flow_mod

LPC_eff_mod

LPC_flow_mod

HPC_eff_mod

HPC_flow_mod

HPT_eff_mod

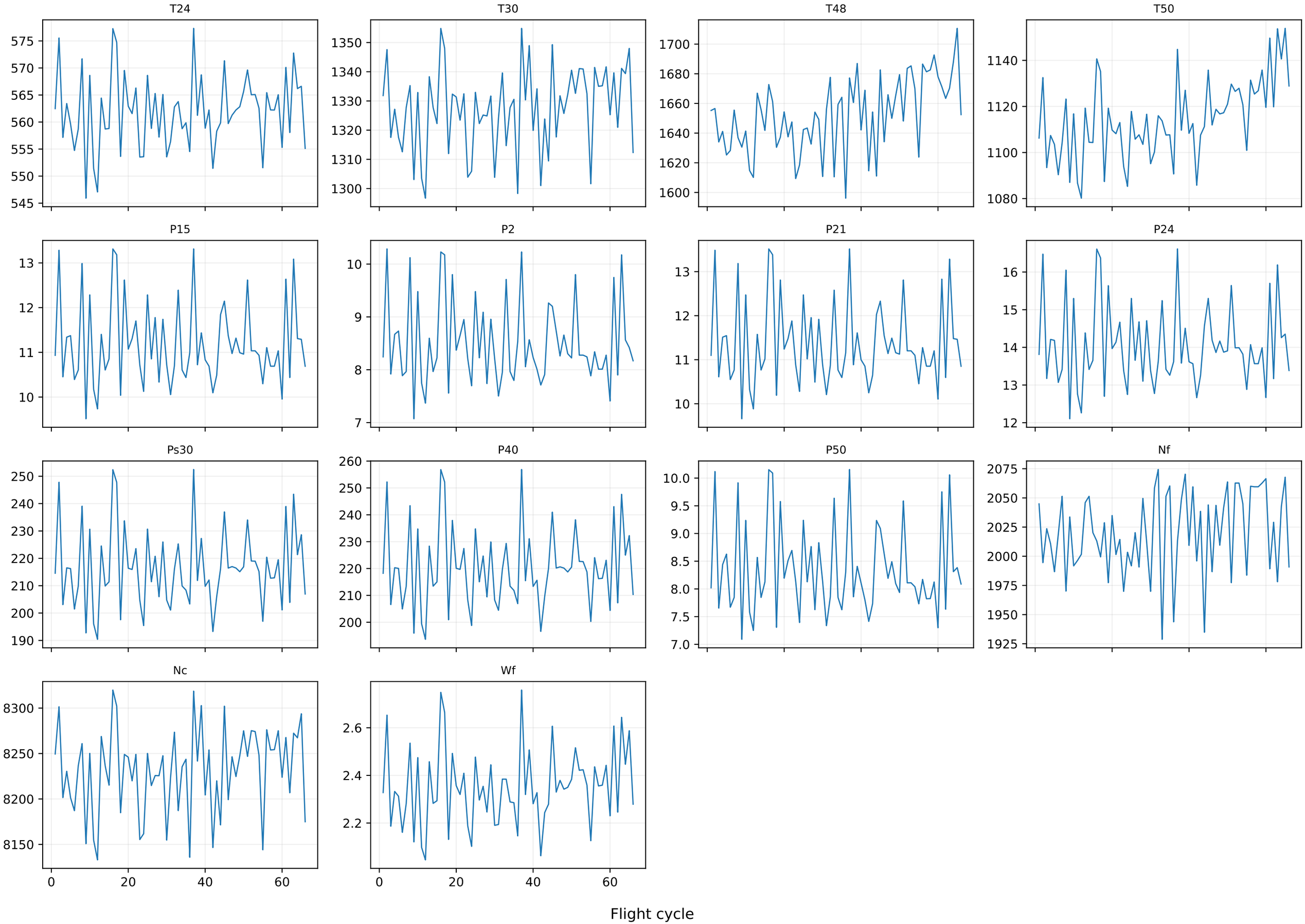
HPT_flow_mod

LPT_eff_mod

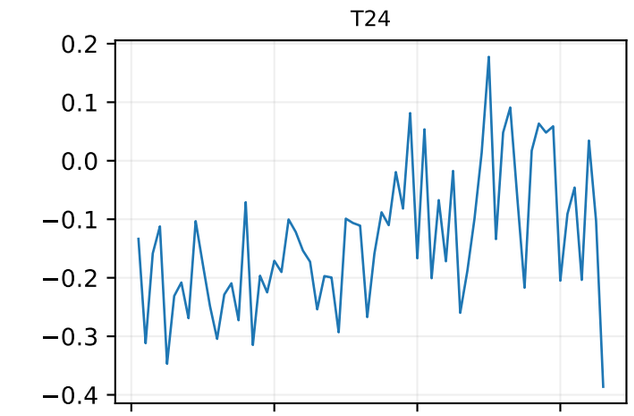
LPT_flow_mod

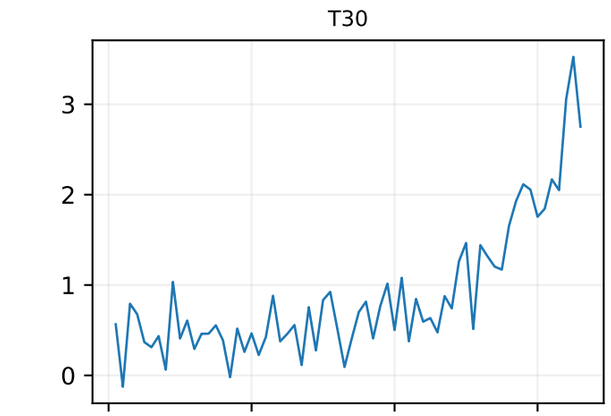
Flight cycle

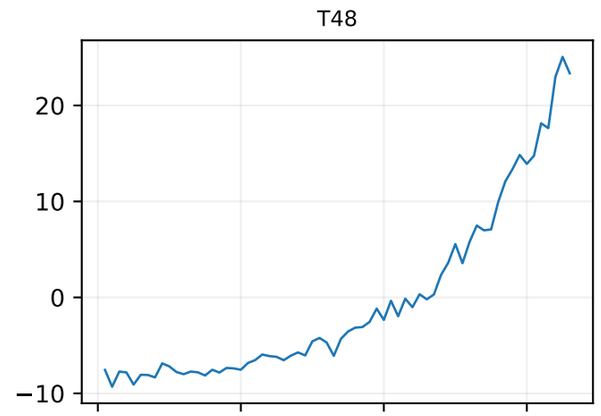
Motor 20: sensors; flight means of selected windows

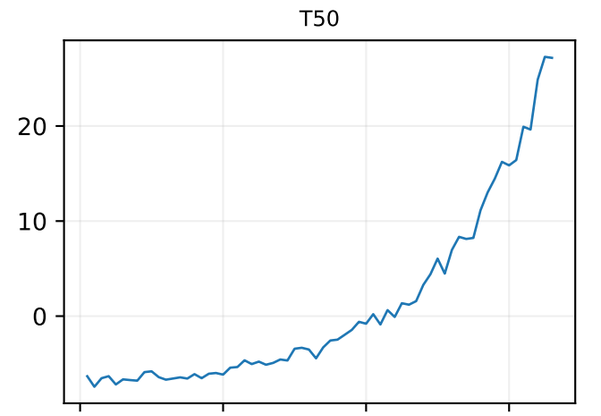


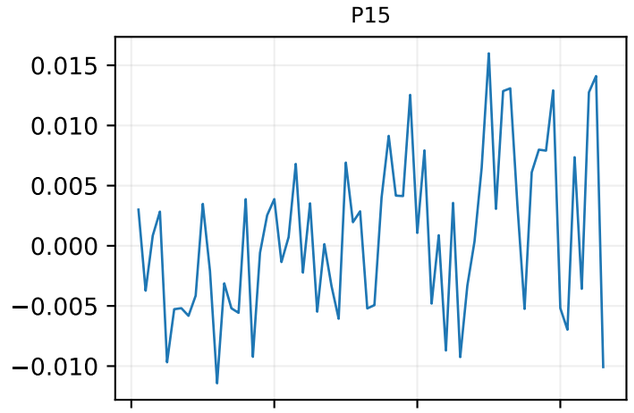
Motor 20: condition_residuals; flight means of selected windows

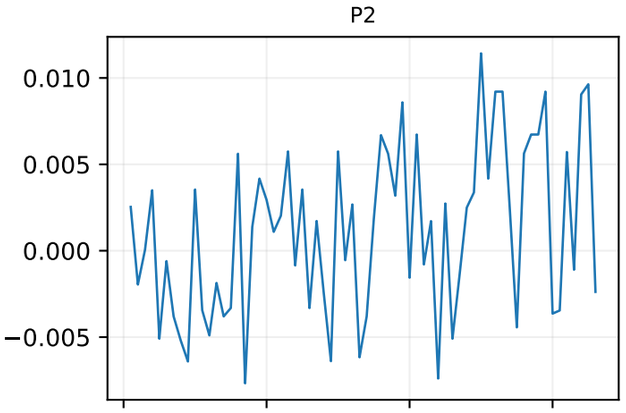


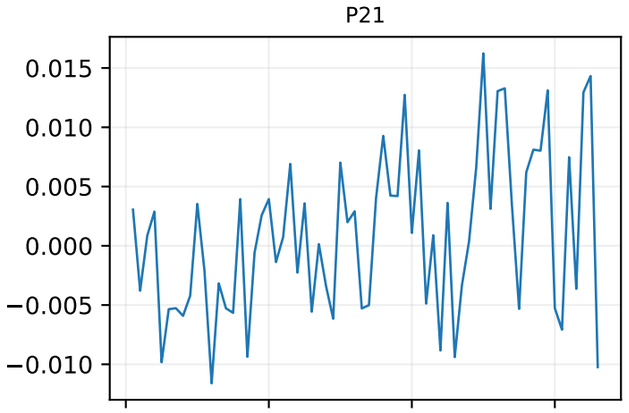


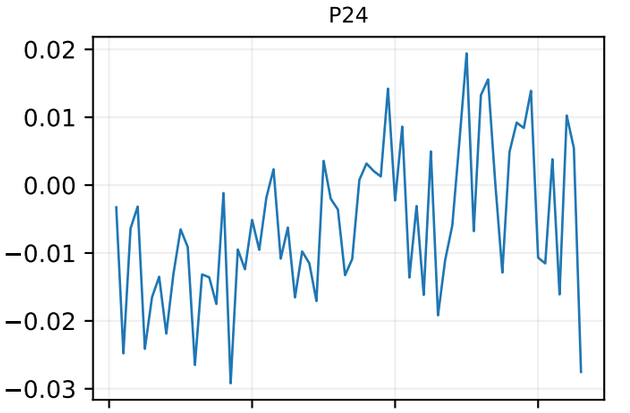


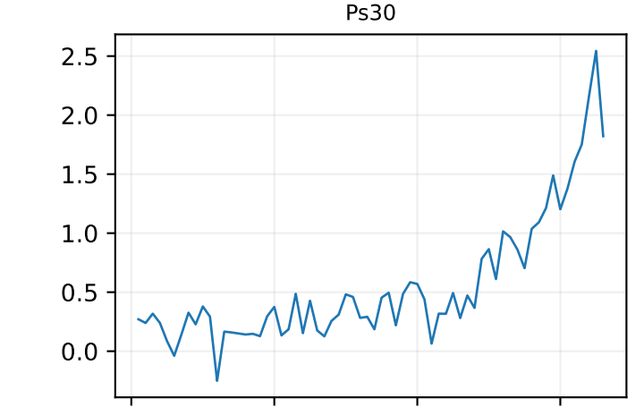


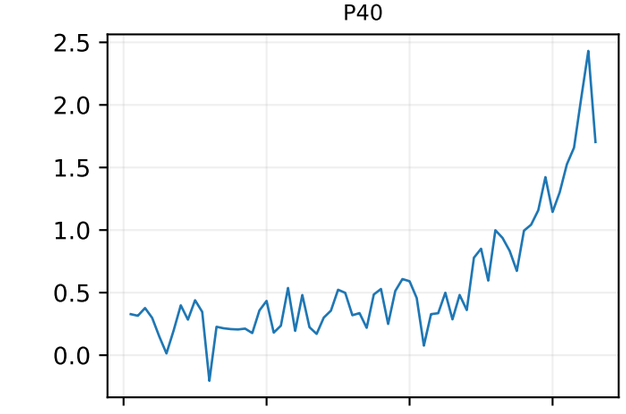


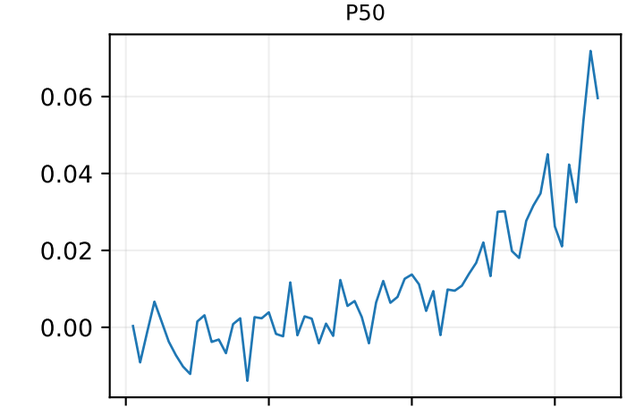


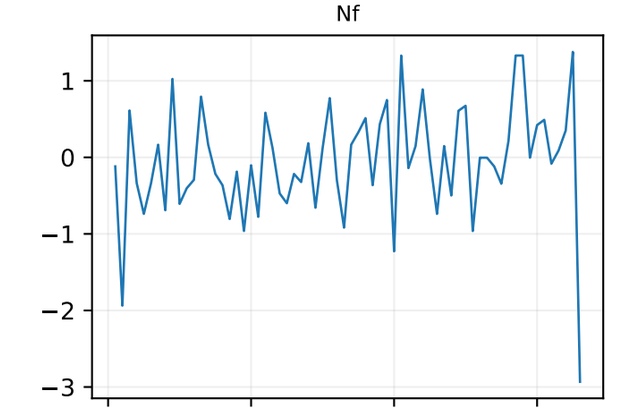


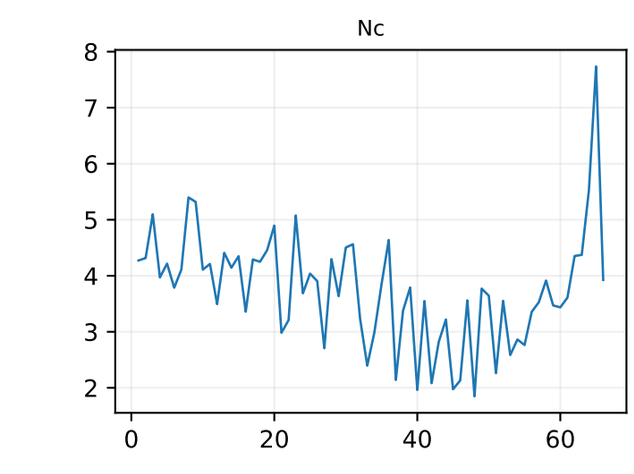


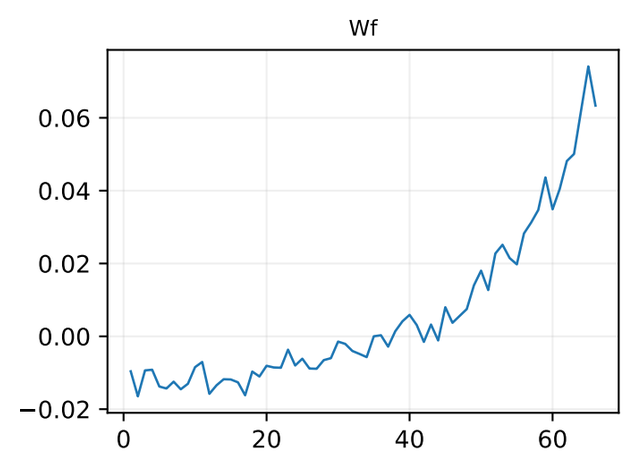












Flight cycle